GEOMETRIC SERIES AND SAVING ON ONE PAGE

£1000 a month to a billion

A savings account is a finite geometric series, and the interest rate decides whether the wait is a lifetime or a geological age.

THE RULE

Multiply by the ratio and subtract

For $S_n = a + ar + ar^2 + \cdots + ar^{n-1}$, the shifted copy rS_n shares every term except the first and last, so $S_n - rS_n = a - ar^n$, and when $|r| < 1$ the last term dies away:

$$S_n = a\,\frac{r^n - 1}{r - 1}, \qquad S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Depositing D at the end of every month into an account paying a nominal annual rate r, credited monthly at $i = r/12$, makes the oldest deposit grow by $(1+i)^{n-1}$ and the newest not at all, so after n months the balance is a finite geometric series with ratio $1+i$:

$$B_n = D\big(1 + (1+i) + \cdots + (1+i)^{n-1}\big) = D\,\frac{(1+i)^n - 1}{i}.$$

ANNUALISING

Nominal, effective and geometric

A quoted 7% credited monthly is a nominal rate. Twelve months of $\frac{7}{12}\%$ compound to the effective annual rate, the AER on a UK savings advert,

$$(1 + \tfrac{r}{12})^{12} - 1 = 7.229\%,$$

and as the crediting becomes continuous the limit is $\mathrm{e}^r - 1 = 7.251\%$. When yearly returns vary, the rate that reproduces the final balance is their geometric mean, which falls below their arithmetic mean by roughly half the variance.

THE INFINITE SERIES

What a monthly £1000 is worth today

Read backwards, discounting each future payment by $\frac{1}{1+i}$ gives an infinite series with ratio below 1. An income of £1000 a month for ever, at 5% nominal, is worth

$$\sum_{k \ge 1} \frac{1000}{(1+i)^k} = \frac{1000}{i} = £240\,000$$

today, the price of a perpetuity; the same sum running forward never converges, which is why a savings balance can grow without limit.

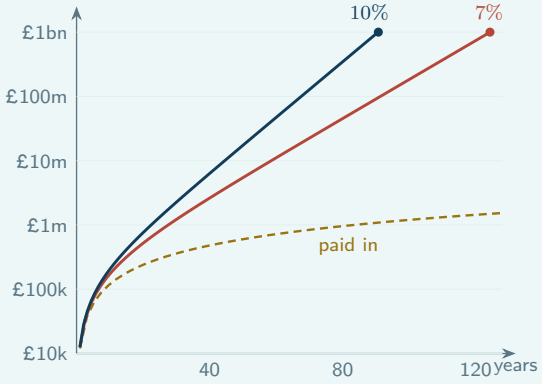
HOW LONG TO A BILLION

Solve the series for n with a logarithm

Setting $B_n = £10^9$ with $D = £1000$ and taking logarithms,

$$n = \frac{\ln\big(1 + 10^9\, i/D\big)}{\ln(1+i)} \text{ months}$$

nominal r	AER	months	years	paid in
0%	0%	1 000 000	83 333	£1000m
3%	3.04%	3134	261	£3.13m
5%	5.12%	2005	167	£2.01m
7%	7.23%	1491	124	£1.49m
10%	10.47%	1088	91	£1.09m
15%	16.08%	759	63	£0.76m



Balance on a logarithmic scale: the sum paid in flattens out while compound interest keeps climbing.

A WORKING LIFETIME

Forty years at 7%

After 480 months the balance is $1000 \times \frac{1.005833^{480} - 1}{0.005833} \approx £2.62\text{m}$ from £480 000 paid in. Money at 7% doubles in about $\ln 2 / \ln 1.07 \approx 10.2$ years, the rule of 72, so each extra decade roughly doubles the pot, which is why the last stretch to a billion is so much faster than the first million.

WHAT THE ARITHMETIC LEAVES OUT

Inflation, volatility and tax

A billion pounds in 124 years' time buys far less than today; at 2% inflation the real rate is about 5%, so a billion in today's money takes about 167 years. Variable investment returns reach a lower final balance than a steady account with the same average, and tax takes a share of the interest along the way. The sums illustrate geometric series and are not financial advice.

Conventions. r is the nominal annual rate, $i = r/12$ the monthly rate, deposits are made at the end of each month, and AER is the annual equivalent rate $(1+i)^{12} - 1$.

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